Dr. Shawn Dsouza

Postdoctoral Researcher in Behavioural & Evolutionary Ecology
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Research Profile

Quantitative evolutionary ecologist and behavioural biologist investigating how human activities reshape animal behaviour, cognitive buffering, and ecosystem dynamics across multiple scales. Methodological expertise spans large-scale meta-analyses, Bayesian phylogenetic hierarchical models, agent-based modelling, and empirical field experimentation in marine and terrestrial ecosystems.

Education

Ph.D. in Biological Sciences — Indian Institute of Science (IISc), Bengaluru, India — **2021 – 2026**
Dissertation: Understanding the scale, mechanisms, and drivers of animal responses to human interactions
Advisor: Prof. Kartik Shanker | Defence completed: 30 June 2026

M.Sc. in Biodiversity — M.E.S. Abasaheb Garware College, Savitribai Phule Pune Univ., India — **2015 – 2017**
Thesis: A cross-taxonomic comparison of bird and butterfly communities of Tamhini Wildlife Sanctuary, Maharashtra, India

B.Sc. in Zoology — S.I.E.S. College of Arts, Science and Commerce, Univ. of Mumbai, India — **2012 – 2015**

Research Experience

Postdoctoral Researcher — NCBS, Bengaluru, India (CaFE Lab; PI: Dr. Meghna Krishnadas) — **Sep 2026 – Present**

- Using long-term monitoring and plant trait data to determine how trait–environment interactions shape seedling survival and community assembly across environmental and disturbance gradients.

Doctoral Researcher — Centre for Ecological Sciences, IISc, Bengaluru, India — **2021 – 2026**

- Synthesised three decades of animal behavioural responses to human super-predators using multi-level meta-analysis (*Ecology Letters*, 2025).
- Developed agent-based models simulating population-level impacts and species coexistence under varying harvest regimes.
- Led field manipulations in the South Andaman Islands evaluating predator removal effects using 3D-printed predator decoys and reef community monitoring.

Research Affiliate — Dakshin Foundation, Bengaluru, India — **May 2022 – Dec 2024**

- Led the elasmobranch monitoring programme in Malvan, Maharashtra, overseeing data collection across multiple landing centres; consulted on marine and fisheries projects.

Research Assistant — Dakshin Foundation, Bengaluru, India — **Jun 2017 – Feb 2021**

- Evaluated fisheries impacts and bycatch resilience on sea snake populations along the Konkan coast using stable isotope analysis and fisheries-dependent methods; managed the organisation's bycatch projects and related outreach.

Grants, Fellowships & Honors

- **Prime Minister's Research Fellowship (PMRF)**, IISc: India's most selective national doctoral fellowship (2022 – 2026).
- **ISBE Travel Grants**, International Society for Behavioural Ecology: $1,600 (2026); $2,500 (2024).
- **GATE Fellowship**, IISc (2021).

- **Independent small research grants as sole PI** (total approx. EUR 10,000): Mangrove Foundation of Maharashtra (2018); The Rufford Foundation, UK (2018); WWF-India (2017).
- **Under review:** Horizon Europe MSCA Postdoctoral Fellowship (submitted 2026), *URBANBRAINS – Cognitive buffering and urban tolerance in birds* (Host: CREAF, Spain; Supervisor: Dr. Daniel Sol).

Publications

Totals: 10 peer-reviewed articles (5 as first author, 1 in press), 2 preprints in review, 3 popular articles. Citations: 60; h-index: 4 (Google Scholar, October 2026).
Note: Asterisk (*) indicates lead / first author.

Peer-Reviewed Journal Articles

1. **Shawn Dsouza*** (2026). Multi-trophic risk from human superpredators may alter predator–prey coexistence and population dynamics. *Individual-based Ecology*, e211852. In press.
2. Patre, M.V., Bora, G., Jakhete, S., **Dsouza, S.**, Shanker, K. (2026). Microplastic accumulation in marine organisms across trophic levels along the west coast of India. *Marine Pollution Bulletin*, 222, 118816.
3. **Shawn Dsouza***, Kartik Shanker, Maria Thaker (2025). Are Human Super-Predators Always Super-Scary? A Meta-Analysis of Wild Animal Behavioural Responses to Human Interactions. *Ecology Letters*, 28(12), e70287.
4. Ruidas, H., Bora, G., **Dsouza, S.**, Shanker, K. (2025). Assessing the impact of microplastics on gonadal health of the spadenose shark (*Scoliodon laticaudus*) on the west coast of India. *Marine Pollution Bulletin*, 212, 117464.
5. Bora, G., **Dsouza, S.**, Shanker, K. (2024). Diet composition and variation in four commonly landed and threatened shark species in Maharashtra, India. *Regional Studies in Marine Science*, 74, 103531.
6. **Shawn Dsouza***, Padhye, A. (2022). A cross-taxonomic comparison of bird and butterfly communities of Tamhini Wildlife Sanctuary, Maharashtra, India, spanning two decades. *Current Science*, 123(10), 1253–1258.
7. **Shawn Dsouza***, Rao, C. (2021). Demographics and reproductive biology of *Hydrophis schistosus* may make it more resilient to bycatch effects than other sea snakes. *Regional Studies in Marine Science*, 47, 101948.
8. **Shawn Dsouza***, Rao, C., Manoharakrishnan, M., Shanker, K. (2021). Resource use and the impacts of fisheries on two sympatric sea snake species on the west coast of India. *Marine Biology*, 168(9), 1–11.
9. Rao, C., **Dsouza, S.**, Gupta, T., Manoharakrishnan, M., Lobo, A.S. (2021). Fisheries induced shift in sea snake community assemblages along the Konkan coast, India. *Aquatic Conservation: Marine and Freshwater Ecosystems*, 31(9), 2402–2411.
10. Jojan, A., **Dsouza, S.**, Mukerjee, P., Rao, C., Shanker, K. (2018). Critiquing narrow critiques of the Convention on Biological Diversity. *Economic & Political Weekly*, 53(44), 17.

Preprints in Review

1. **Shawn Dsouza***, Joshi, A., Thaker, M., Shanker, K. (2025). Out of sight, out of mind: Removal of predators by fisheries alters reef fish behaviour. *bioRxiv*.
2. **Shawn Dsouza***, Joshi, A., Ahuja, B., Ahmad, S., Shanker, K. (2025). Local extraction of predatory fish may initiate behaviourally mediated trophic cascades in coral reefs in the Andaman Islands. *bioRxiv*.

Popular Articles

- Dsouza, S. (2020). Locked down in a fishing town. *The Bastion*.
- Dsouza, S. (2019). Scaly business – a day in the life of a sea snake ecologist. *Current Conservation*, 13.2.
- Creasey, M., Dsouza, S. (2019). Sailing into an uncertain future. *Current Conservation*, 13.3.

Conference Presentations

- Poster: Are human super-predators always super-scary? A meta-analysis of wild animal behavioural responses to human interactions. GRS Predator–Prey Interactions Seminar, Barga (2026).

- Poster: Local extraction of predatory fish may initiate behaviourally mediated trophic cascades in coral reefs in the Andaman Islands. GRC Predator–Prey Interactions Meeting, Barga (2026).
- Talks: Are super-predators always super-scary? & Fishes do fear the reaper: marine protected areas alter fish assemblages and behaviour. Behaviour 2025, Kolkata (2025).
- Talk: Are superpredators also super-scary? A systematic review and meta-analysis of the effect of human interactions on animal behaviour. ISBE Congress, Melbourne (2024).
- Invited talk: Sea snakes: ecology and threats. University of Exeter (2019).
- Student talk: Fishing for sea snakes. SCCS Cambridge (2019). Student poster: Can butterflies improve conservation? SCCS Bangalore (2017).

Professional Service & Memberships

- Symposium organiser: "Global Change Impacts on Predator–Prey Ecology and Evolution", GRS Predator–Prey Interactions Seminar, Barga (2026).
- Symposium organiser: "Responses and adaptations to human super-predators and anthropogenically modified environments", Behaviour 2025, Kolkata (co-chairs: Chayan Munshi, Maria Thaker).
- Member: International Society for Behavioral Ecology (ISBE, since 2023); Association for the Study of Animal Behaviour (ASAB, since 2022).

Mentorship & Teaching

- Mentored 6 Master's thesis students at IISc to completion; 3 have progressed to doctoral programmes and all 6 have co-authored peer-reviewed publications.

Technical & Methodological Skills

- **Statistical Modelling & Programming:** R, Python, Bayesian multilevel inference (`brms`, `Stan`, `MCMCglmm`), agent-based modelling, phylogenetic comparative methods (PGLMM, confirmatory path analysis).
- **Evidence Synthesis & Workflows:** Multi-level meta-analysis, PRISMA systematic reviews, large-scale secondary database harmonisation, Git/GitHub, Docker, LaTeX, Linux-based reproducible pipelines.
- **Fieldwork & Experimental Design:** Coral reef visual census, baited remote underwater video surveys (BRUVS), 3D-printed experimental decoys, stable isotope–based food-web analysis, fisheries-dependent sampling, point counts & line transects.

Referees

Prof. Kartik Shanker, Centre for Ecological Sciences, IISc, Bengaluru (kshanker@iisc.ac.in) | Prof. Maria Thaker, Centre for Ecological Sciences, IISc, Bengaluru (mthaker@iisc.ac.in)